

Chapter

Microalgae and Cyanobacteria in Space: Current Trends, Challenges and Prospects for Space Biotechnology

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Abstract

Microalgae and cyanobacteria are emerging as foundational biological resources for sustainable space exploration and long-duration human missions. Their high photosynthetic efficiency, rapid growth, resilience to diverse environmental stresses, and ability to simultaneously fix carbon dioxide (CO₂), produce oxygen (O₂), and generate biomass make them promising components of bioregenerative life-support systems (BLSS). This chapter provides an overview of historical and contemporary spaceflight experiments involving microalgae and cyanobacteria, highlighting their physiological responses to space conditions and their potential roles in closed-loop life-support systems. Recent advances in cultivation technologies, omics-based approaches, and integrated bioprocessing platforms are discussed, with emphasis on applications in air revitalization, wastewater treatment, nutrient recycling, food production, and biomanufacturing. The major challenges associated with extraterrestrial cultivation, including microgravity, radiation exposure, limited resources, environmental instability, and biosafety, are examined together with emerging strategies to improve system reliability and biological performance. Advances in genetic engineering, synthetic biology, artificial intelligence (AI)-assisted modeling, automation, and process optimization offer new opportunities to develop robust and multifunctional biological systems for space applications. The chapter further explores future prospects for engineered, highly resilient strains and integrated space biorefineries capable of converting waste streams and available resources into oxygen, food, fuels, biomaterials, and other valuable products. Potential applications beyond low-Earth orbit, including lunar and Martian habitats, in situ resource utilization, and long-term planetary ecosystem development, are also considered. Overall, microalgae and cyanobacteria represent versatile biological platforms with significant potential to support sustainable, resource-efficient, and increasingly autonomous human activities beyond Earth.

Keywords: microalgae, cyanobacteria, space biotechnology, bioregenerative life-support systems, photobioreactors, synthetic biology, space biomanufacturing

1. Introduction

Space exploration is shifting from short orbital missions to long-duration expeditions, lunar bases, and future human settlement on Mars [1]. As mission duration increases, dependence on frequent resupply from Earth becomes impractical and costly, especially for water, oxygen, food, and other consumables [2, 3]. Consequently, autonomous and regenerative life-support systems are essential for sustainable space habitation [4]. Bioregenerative life-support systems (BLSS) are considered key components of future long-term missions [5]. Unlike physico-chemical systems, which depend on consumables and high energy inputs, BLSS use living organisms to recycle resources and close material loops [6]. These systems can capture carbon dioxide, generate oxygen, recycle wastewater, recover nutrients, and produce edible biomass with minimal external inputs [7]. Such capabilities are particularly valuable for lunar habitats, Martian settlements, and deep-space travel [8].

Among biological candidates, microalgae and cyanobacteria are especially promising due to their high photosynthetic efficiency, rapid growth, and metabolic flexibility [9, 10]. They can tolerate environmental stresses relevant to spaceflight, including radiation, nutrient limitation, oxidative stress, and fluctuating light conditions [11]. Studies have also shown that several species can remain metabolically active under microgravity and survive extreme extraterrestrial conditions [12]. In addition to life-support roles, their biomass provides proteins, polyunsaturated fatty acids (PUFAs), vitamins, pigments, and antioxidants beneficial for astronaut nutrition and health [13]. Through metabolic engineering, they may also produce pharmaceuticals, nutraceuticals, biopolymers, and radiation-protective compounds, supporting a future space bioeconomy [14].

As illustrated in **Figure 1**, the conceptual diagram of an algal-based BLSS highlights CO₂ capture from crew respiration, O₂ generation via photosynthesis, nutrient recycling from wastewater and human waste streams, and biomass harvesting for food production, bioactive compounds, and biofuel applications. Therefore, this chapter reviews current advances and future prospects in space algal biotechnology. It summarizes the biological potential of microalgae and cyanobacteria, historical and recent spaceflight experiments, and their responses to microgravity, radiation, and extraterrestrial resource limitations. It also discusses innovations in photobioreactor (PBR) design, synthetic biology, and in situ resource utilization (ISRU), while addressing operational challenges such as biofilm formation and gas–liquid mass transfer limitations. Overall, the chapter highlights the role of algae-centered systems in enabling sustainable human exploration beyond Earth.

2. Microalgae as key organisms for space applications

2.1 Biological features ideal for space environments

Microalgae and cyanobacteria possess traits that make them highly suitable for extraterrestrial deployment, offering advantages over higher plants in closed habitats [10]. Species such as *Chlorella vulgaris*, *Arthrospira platensis* (Spirulina), and *Synechococcus elongatus* can double their biomass within 12–24 h, enabling rapid and continuous production of food, oxygen, and useful bioproducts within limited



Figure 1.
 Conceptual diagram of an algal-based bioregenerative life-support system (BLSS).

spacecraft volume [10, 15]. Their fast growth also improves resilience, as cultures can recover quickly after contamination or failure. Many species are metabolically flexible and can grow mixotrophically using light plus organic carbon sources derived from waste streams, helping maintain productivity during low-light conditions such as lunar nights or Martian dust storms [16]. Extremophilic taxa, including *Chroococcidiopsis*, tolerate desiccation, radiation, and extreme temperatures, supporting their use in harsh planetary environments [17, 18]. Microalgae also exhibit high photosynthetic efficiency and convert most of their biomass into edible or usable material, unlike crops that invest heavily in roots and stems [19]. Their unicellular structure allows precise cultivation in PBRs, where light, nutrients, gas exchange, and harvesting can be tightly controlled with minimal crew labor [15]. Importantly, microalgae perform dual life-support roles by fixing CO_2 and generating O_2 , helping stabilize cabin atmosphere while producing biomass for food and manufacturing [20]. This makes them valuable for regenerative systems where mass efficiency and reliability are essential.

2.2 Nutritional and biochemical advantages

Microalgae are highly nutrient-dense and well suited for in situ food production during deep-space missions [13]. Many species contain 40–70% protein by dry weight, exceeding common terrestrial crops, and often provide balanced amino acid profiles important for preserving muscle mass in microgravity [11]. They are also rich in beneficial lipids, especially omega-3 fatty acids such as EPA and DHA, which support cardiovascular, neurological, and immune health [21]. In addition, microalgae supply vitamins, provitamin A, and trace minerals, offering a fresh

alternative to stored supplements that may degrade during long missions [22]. Many species naturally produce antioxidants and radioprotective compounds. Examples include phycocyanin from *Spirulina* and astaxanthin from *Haematococcus pluvialis*, which may help reduce oxidative stress and radiation damage during spaceflight [23]. Algal biomass can also be processed into protein isolates, nutraceuticals, and palatable foods through fermentation or 3D food printing, helping improve dietary variety and crew morale [24].

2.3 Genetically engineered and synthetic biology approaches

Genetic engineering can greatly expand the utility of microalgae for space systems [25]. Tools such as CRISPR–Cas, TALENs, and zinc-finger nucleases enable precise modification of genes related to stress tolerance, photosynthesis, and metabolite production [26]. For example, truncated light-harvesting antenna mutants reduce shading in dense cultures, improving light penetration and biomass productivity in compact PBRs [27]. Other engineering targets include enhanced DNA repair, antioxidant defenses, and protective pigment production to improve radiation resistance [28, 29]. Synthetic biology also allows microalgae to function as biofactories, producing vaccines, antibodies, pharmaceuticals, nutraceuticals, and biopolymers using sunlight and recycled CO₂ [30]. This could reduce dependence on Earth resupply while supporting crew health and material needs. Overall, engineered microalgae represent multifunctional platforms for food production, environmental control, and in situ biomanufacturing, making them key technologies for sustainable long-duration human exploration beyond Earth.

3. Historical and modern spaceflight experiments

3.1 Early experiments

Table 1 summarizes the historical and experimental context of key space microalgae experiments. Space-based algal biotechnology began with pioneering

Experiment/ program	Species	Duration	Key findings	References
BIOS-3	<i>Chlorella vulgaris</i>	1–6 months	Demonstrated oxygen regeneration and CO ₂ fixation in a closed environment	Wheeler [31]
MELiSSA	<i>Chlorella</i> , consortia	Multi-compartment loop	Integration of an algal compartment into waste recycling and oxygen production	Lasseur et al. [35]
ISS Growth Studies	<i>Spirulina platensis</i> , <i>Chlorella</i>	Days to weeks	Altered growth kinetics and pigment composition under microgravity	Fahrion et al. [36]
Omics-level flight experiments	<i>Arthrospira platensis</i>	Short-term	Upregulation of stress genes and pigment accumulation	Fisher et al. [37]

Table 1.
Summary of key space microalgae experiments.

studies in the mid-twentieth century, particularly the Soviet BIOS-3 program at the Institute of Biophysics in Krasnoyarsk [31]. BIOS-3 was one of the earliest Closed Ecological Life Support System (CELSS) prototypes designed to sustain human life in a sealed environment using phototrophic microorganisms [32]. *Chlorella* species were selected because of their rapid growth, strong photosynthetic performance, and high oxygen production capacity [33]. Ground-based BIOS-3 trials, some lasting up to six months with human crews, demonstrated that microalgae could regenerate oxygen, remove carbon dioxide, and provide nutritional biomass within closed habitats [34]. These studies offered the first strong proof-of-concept for BLSS. However, BIOS-3 also revealed the difficulty of maintaining stable closed biological loops. Small imbalances between algal gas exchange and crew respiration caused fluctuations in O₂ and CO₂ levels, highlighting the need for precise environmental control and continuous monitoring [33].

In parallel, the European Space Agency developed the Micro-Ecological Life Support System Alternative (MELiSSA), which adopted a modular and compartmentalized design [35]. Instead of a single closed unit, MELiSSA used linked microbial and phototrophic compartments to convert waste into oxygen, nutrients, and edible biomass. Its algal module commonly employed *Limnospira indica* (formerly *Spirulina*) for oxygen production, carbon fixation, and biomass generation [38]. Together, BIOS-3 and MELiSSA confirmed that phototrophic microorganisms can perform essential life-support functions in closed systems. They also showed that compartmentalization improves resilience, process control, and system stability [36, 39]. These early programs laid the foundation for later experiments aboard the International Space Station (ISS) and for future lunar and Martian habitat design.

3.2 Experiments on the ISS

Research aboard the ISS has greatly improved understanding of microalgal physiology under microgravity and space radiation [40, 41]. Many species maintain active growth in orbit, although others show changes in cell morphology, pigment composition, and division rates, indicating strain-specific responses to the space environment [42]. A major challenge in microgravity is the absence of buoyancy-driven convection. On Earth, natural mixing helps distribute nutrients and remove waste products. In orbit, transport becomes diffusion-limited, creating localized nutrient depletion and metabolite accumulation around cells [43, 44]. These conditions can reduce culture performance and require specially designed PBRs. To address this, several microgravity-adapted PBR systems have been developed. Membrane-based reactors use diffusive gas exchange instead of bubbling, avoiding foam formation and gas separation problems in weightlessness [45]. Other systems use rotating vessels, microfluidic channels, or mechanical mixing to maintain uniform nutrient distribution and prevent stagnation. Such designs are important for sustaining dense cultures during long missions.

Microgravity also influences photosynthesis. Changes in chloroplast positioning, thylakoid structure, and carbon fixation have been reported when gravitational cues are removed [46]. Some species show reduced photosystem II efficiency, whereas others compensate by increasing pigment synthesis or protective non-photochemical quenching mechanisms [47]. In addition, space radiation causes molecular and metabolic stress. Transcriptomic and proteomic studies show activation of antioxidant

systems, DNA repair pathways, and shifts in carbon and nitrogen metabolism [48, 49]. Energy is often redirected from growth toward maintenance and survival. Overall, ISS experiments have clarified how microgravity and radiation affect algal performance. These findings provide valuable guidance for selecting resilient strains and designing next-generation PBRs for BLSS in lunar, Martian, and deep-space missions.

3.3 Omics-level findings

High-throughput omics technologies have greatly improved understanding of how microalgae adapt to spaceflight conditions [25]. Transcriptomic studies commonly show upregulation of stress-response genes related to redox balance, lipid remodeling, protein folding, and chaperone activity [50]. This suggests that cells shift resources from growth toward protection and repair. Increased expression of heat shock proteins (HSPs) further indicates that microgravity induces proteotoxic stress requiring continuous protein stabilization [51].

Metabolomic and lipidomic analyses reveal accumulation of antioxidant and photoprotective compounds such as astaxanthin, β -carotene, and tocopherols, which help scavenge reactive oxygen species (ROS) and protect DNA, membranes, and photosynthetic machinery from radiation-induced damage [52]. Lipid remodeling also helps maintain membrane fluidity, nutrient transport, and cellular signaling under diffusion-limited microgravity conditions [53].

Proteomic studies provide further evidence of coordinated regulation of energy metabolism, photosystem stability, ROS detoxification, and stress-signaling pathways [50]. Together, these findings demonstrate the strong metabolic plasticity of microalgae under space stressors. Importantly, omics data also identify targets for engineering improved strains. Genes controlling antioxidant defenses, DNA repair, and stress tolerance may be exploited to develop space-hardened microalgae with higher resilience and productivity, strengthening future BLSS performance. **Table 2** summarizes observed physiological changes of microalgae under microgravity.

Trait/parameter	Response under microgravity	Mechanistic insight	Implications for BLSS
Growth rate	Variable; sometimes reduced	Altered nutrient/gas diffusion	Requires optimized PBR mixing
Morphology	Cell enlargement or deformation	Cytoskeletal rearrangement	Can affect photosynthesis
Photosynthetic efficiency	Decrease or compensatory increase	Chloroplast repositioning	Adjust light intensity or spectral quality
Pigment accumulation	↑ Carotenoids, phycobiliproteins	Photoprotection and ROS mitigation	Enhances antioxidant content
ROS levels	↑ Oxidative stress	Radiation and microgravity-induced stress	Necessitates antioxidant supplementation

Table 2.
Physiological responses of microalgae under microgravity.

3.4 Limitations of current research

Despite substantial progress, important barriers still limit the translation of experimental results into fully operational BLSS [54]. Most spaceflight studies use milliliter-scale cuvettes, culture bags, or small PBRs that do not reflect the large volumes required for real missions [36, 44]. Supporting even a small crew would require hundreds to thousands of liters of culture, creating additional challenges in mixing, light penetration, gas–liquid mass transfer, and nutrient distribution. Environmental conditions aboard the ISS are also limited. Constraints in light quality, irradiance uniformity, gas exchange, power supply, and thermal control often prevent testing under fully optimized operating conditions [20, 55]. As a result, current studies may underestimate the true productivity of microalgae in future dedicated habitat systems [56].

Another major gap is long-term ecological stability. Many experiments are short-duration and conducted under near-sterile conditions to avoid contamination [48, 57]. Therefore, risks such as phage infection, bacterial overgrowth, community shifts, inhibitory metabolite buildup, and sudden culture collapse remain poorly understood. In a closed-loop BLSS, a culture crash could disrupt oxygen production, carbon dioxide removal, and food supply simultaneously, posing serious risks to crew safety [58]. To overcome these limitations, future research must focus on long-duration, integrated system trials that include automated harvesting, biomass processing, reinoculation strategies, and real-time monitoring [59]. Multi-year validation at realistic scales will be essential to demonstrate the reliability, controllability, and safety of algal-based BLSS for sustainable human presence beyond Earth.

3.5 Implications for future missions

Findings from BIOS-3, MELiSSA, and ISS experiments demonstrate both the strong potential and operational complexity of using microalgae in long-duration space missions [16, 40, 59]. These studies confirm that microalgae can provide key life-support functions, including oxygen production, carbon dioxide removal, and biomass generation, when environmental conditions are carefully controlled. However, scaling from laboratory cultures to systems capable of supporting human crews presents major engineering and biological challenges. These include maintaining efficient gas–liquid mass transfer, preventing contamination, ensuring uniform light delivery, and preserving metabolic stability under prolonged microgravity and radiation exposure [36].

Insights from multi-omics research are especially valuable for future applications, as they help identify traits linked to antioxidant capacity, photoprotection, stress resistance, and efficient resource utilization. Such knowledge can guide strain selection and genetic improvement for more reliable space cultivation systems [25, 50]. Future lunar bases, Martian habitats, and deep-space vehicles will likely require integrated and compartmentalized life-support systems rather than single biological units [60]. These systems must combine robust algal modules with advanced sensors, automation, and adaptive control to maintain long-term stability and crew safety.

4. Algal biotechnology for bioregenerative life-support systems

4.1 Air revitalization

Microalgae are key components of BLSS because of their strong ability to regenerate cabin atmospheres by capturing CO₂ and releasing O₂ through photosynthesis [20, 39]. Unlike physicochemical systems used on the ISS, which require consumables or energy-intensive regeneration, algal systems are self-renewing and operate mainly using light and recycled nutrients [5]. Their high photosynthetic efficiency often exceeds that of higher plants when compared by biomass or illuminated area, largely because they lack non-productive tissues such as roots and stems [10]. Species such as *Chlorella vulgaris* and *Arthrospira platensis* (Spirulina) can maintain stable oxygen production under controlled conditions, even in microgravity [37, 44].

To overcome reduced natural mixing in space, cultivation systems use mechanical stirring, pneumatic circulation, or membrane-based gas exchange to distribute CO₂ and remove O₂ efficiently [61]. Photosynthetic activity can also be rapidly adjusted through light control, allowing microalgae to respond quickly to temporary increases in crew CO₂ production [62]. In addition, dense liquid algal cultures can reduce equivalent system mass (ESM) compared with crop-based systems, saving launch mass, habitat volume, and crew time [63]. These advantages make microalgae highly effective for air revitalization in long-duration lunar, Martian, and deep-space missions.

4.2 Wastewater and nutrient recycling

In BLSS, microalgae play a dual role as waste-treatment agents and nutrient-recovery systems [20]. They efficiently absorb nitrogen and phosphorus from human waste streams such as treated urine and processed fecal matter, converting these wastes into new biomass [64]. Nitrogen recovery is especially important because urine contains high ammonium levels, which can become toxic to some species. This can be managed through bacterial nitrification, which converts ammonium to nitrate, or by using tolerant strains such as *Chlorella* spp. [64]. Microalgae also capture phosphorus, an essential but limited nutrient required for nucleic acids and energy metabolism. Recycling phosphorus within biomass is critical in closed-loop systems where external resupply is difficult [65].

Greywater treatment can be improved using algal–bacterial consortia. Bacteria break down organic compounds into CO₂ and mineral nutrients, while microalgae supply oxygen through photosynthesis and assimilate the released nutrients into biomass [66]. These systems can remove pollutants and microbes while producing useful biomass for food or biomanufacturing. By converting waste into reusable resources, algal systems help close ecological loops, reduce dependence on Earth resupply, and improve the sustainability and resilience of future lunar, Martian, and deep-space habitats [59].

4.3 Food and nutritional supplementation

Edible microalgae such as *Chlorella* and *Arthrospira* (Spirulina) are nutrient-dense foods with major potential for space missions [67]. Their biomass commonly

contains 40–70% protein by dry weight, provides all essential amino acids, and is rich in beneficial PUFAs, including omega-3 DHA and EPA [11, 13]. Microalgae also supply vitamins and antioxidants that may degrade in stored space foods, such as vitamin B12, provitamin A, and carotenoids like astaxanthin. These compounds may help protect astronauts against oxidative stress, radiation damage, and physiological decline during long missions [41, 68]. Producing fresh biomass on-site ensures continued access to these bioactive nutrients [69].

Challenges remain in taste, texture, and digestibility. Some species have rigid cell walls that reduce nutrient absorption, so processing methods such as bead milling, homogenization, or enzymatic hydrolysis are often needed [70]. Advanced food technologies, including algal protein isolates and 3D food printing, can create more familiar and customized meals, helping reduce menu fatigue and improve crew morale [71]. Beyond nutrition, microalgae may also function as “pharmaceutical foods” by delivering immunomodulatory peptides, anti-inflammatory carotenoids, and antioxidant compounds. These properties could help counter spaceflight-related immune dysfunction, bone loss, inflammation, and oxidative stress [72]. Thus, microalgae offer a multifunctional food platform for long-duration BLSS missions.

4.4 Biofuel and bioproduct generation

Beyond air revitalization and food production, microalgae are valuable platforms for producing renewable fuels and high-value materials, supporting sustainable long-duration missions and space settlements [16, 20]. Under stress conditions such as nitrogen limitation, salinity changes, or high light, many species accumulate lipids or carbohydrates that can be converted into biodiesel, bioethanol, and other biofuels [73]. These fuels could support surface vehicles, backup power systems, or habitat heating, reducing dependence on Earth-supplied resources [74]. Some species, including *Chlamydomonas reinhardtii*, can also produce biohydrogen under anaerobic or sulfur-limited conditions, offering a clean energy option for in situ use [75].

Algal biotechnology also supports ISRU through material production. Biomass can be processed into biodegradable polymers such as polyhydroxybutyrate (PHB), which may be used in additive manufacturing or 3D printing of tools, spare parts, and packaging materials [76]. This local production model can reduce payload mass, launch costs, and dependence on resupply missions [77]. In addition, engineered microalgae can produce valuable compounds such as antiviral agents, antibiotics, immunomodulatory peptides, enzymes, and diagnostic reagents using light, CO₂, and recycled nutrients as inputs [40, 50]. Together, these capabilities transform algal systems from simple life-support units into multifunctional platforms for energy generation, manufacturing, healthcare support, and scientific operations, making them central to self-sufficient space habitats.

5. Microalgae–materials interface and space biomanufacturing

5.1 PBR engineering

Successful use of microalgae in space depends on PBRs that maintain high productivity under microgravity, radiation exposure, and limited resources [15, 78]. These systems must provide efficient light delivery, gas exchange, and nutrient distribution without natural convection, which is absent in space and reduced on the

Moon or Mars [36]. Recent designs include capillary-driven and microfluidic PBRs that use engineered channels, porous materials, and surface-tension forces for passive fluid transport [79]. In microgravity, this allows uniform nutrient supply and gas exchange with low energy demand, reduced shear stress, and improved long-term culture stability [80].

Light management is also critical because sunlight may be interrupted by orbital cycles, lunar nights, or Martian dust storms. High-efficiency LED systems are therefore important as primary or supplemental light sources [81]. Red and blue wavelengths are commonly used to maximize photosynthetic efficiency, while pulsed-light strategies can improve energy conversion compared with continuous lighting [82, 83]. Optical fibers, reflective surfaces, and light guides can further improve internal light distribution in dense cultures.

Biofouling is another challenge, as algal or bacterial films on reactor surfaces reduce light penetration and mass transfer [84]. Automated cleaning systems using scraping, fluid jets, or compatible chemical rinses may help control fouling [85]. Gas management is equally important. Membrane-based PBRs using gas-permeable materials such as silicone or fluoropolymers allow efficient CO₂ delivery and O₂ removal by diffusion, avoiding foaming and gas separation problems common with bubble aeration in microgravity [86].

As illustrated in **Figure 2**, PBR designs optimized for microgravity conditions include several space-adapted configurations, such as capillary-driven systems that

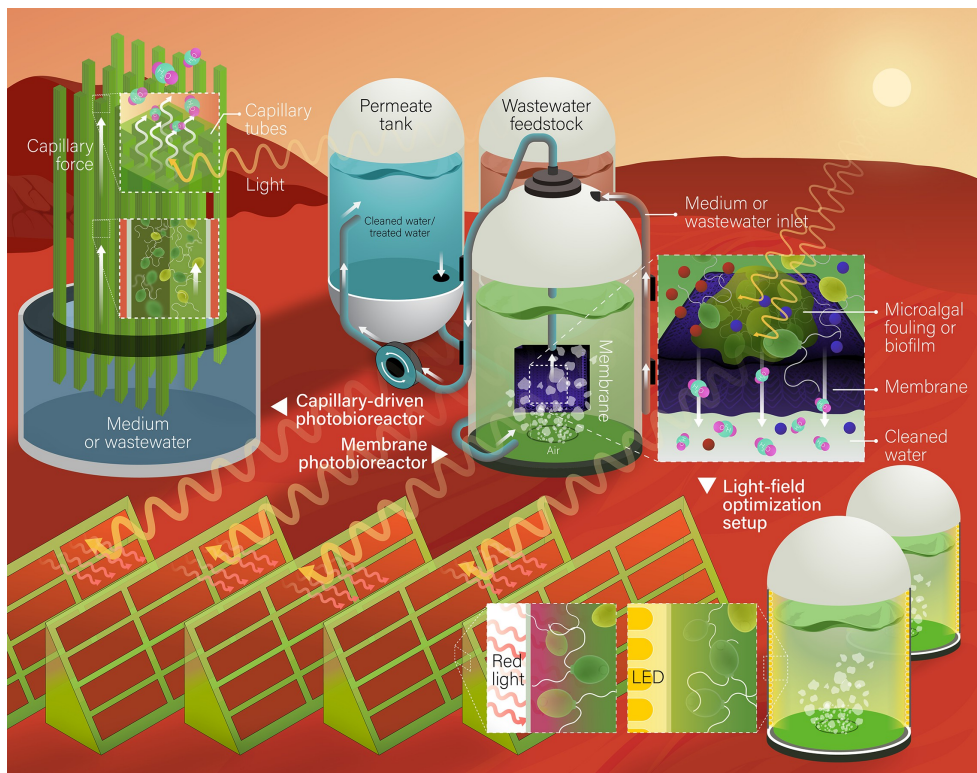


Figure 2.
Photobioreactor designs optimized for microgravity conditions.

enable nutrient and gas transport in the absence of convection, membrane-based bioreactors with selective gas exchange, and light-field optimization strategies, including LED arrays and internal light-scattering structures. Overall, advances in fluidics, lighting, and gas exchange are enabling scalable, low-power, and reliable algal cultivation systems for long-duration missions, planetary habitats, and closed-loop BLSS.

5.2 Biomaterials for ISRU

Microalgal biomass is a versatile feedstock for ISRU, offering a renewable alternative to Earth-supplied construction and consumable materials in space habitats [9]. Its main biopolymers, including starch, cellulose, alginate, and protein-rich fractions, can be processed into films, coatings, and biodegradable products [87]. Some algal taxa also produce hydrocolloids such as alginate and carrageenan, which can be molded or cross-linked into materials with tunable mechanical properties [13]. Material properties such as strength, flexibility, permeability, and degradation rate can be adjusted through processing conditions, enabling application-specific design for space habitats [88]. Further functionality is achieved through hybrid biocomposites that combine algal-derived polymers with in situ materials such as lunar or Martian regolith [89].

In these systems, algal extracellular polymeric substances (EPS) act as natural binders, and when combined with fungal or microbial networks, can form biologically based cement-like materials [90]. These composites can be manufactured using casting, extrusion, or 3D printing, producing structural elements with reduced mass and acceptable mechanical strength for habitat construction [91]. Because regolith can comprise most of the material volume, launch mass requirements are significantly reduced [92]. Beyond structural use, these biomaterials can also provide thermal insulation, water retention, and potentially self-healing properties when biologically active components are included [93, 94]. Overall, microalgal biomaterials support low-mass, sustainable, and adaptable construction strategies for future extraterrestrial habitats.

5.3 Radiation-shielding applications

Radiation is a major hazard for astronauts on the Moon, Mars, and deep-space missions because these environments lack strong atmospheric or magnetic protection from solar particle events (SPEs) and galactic cosmic rays (GCRs) [95, 96]. Microalgae produce photoprotective pigments such as scytonemin and mycosporine-like amino acids (MAAs), which strongly absorb UV radiation and possess antioxidant properties. These compounds can reduce ROS and help limit radiation-induced cellular damage [97]. As renewable biological products, they offer promising protection for both crew health and habitat materials.

Algal pigments and biomass have also been proposed as components of shielding materials. Layered systems made from living cultures or processed algal biomass may provide partial radiation attenuation while simultaneously supporting oxygen production, CO₂ capture, and nutrient generation [59, 98]. The high water content and the abundance of light elements such as hydrogen, carbon, and oxygen in algal biomass can help slow and scatter high-energy charged particles, potentially offering a mass-efficient alternative to traditional shielding materials like aluminum [99]. This approach reflects a multifunctional habitat design, where biomass generated from food

production, air revitalization, or waste recycling is reused as a shielding material. Such integration improves resource efficiency and supports resilient, closed-loop space habitats for long-duration human exploration beyond Earth.

5.4 Challenges and constraints

5.4.1 Microgravity effects

Microgravity strongly influences microalgal cultivation by altering fluid behavior and mass transfer. Without buoyancy-driven convection, nutrient and gas movement depends mainly on molecular diffusion, which is slower and less efficient in PBRs [99, 100]. This can create localized nutrient depletion around cells and oxygen accumulation within cultures. High dissolved oxygen levels may cause photo-oxidative stress, inhibit photosystem II, and reduce photosynthetic efficiency, lowering biomass productivity [101]. Dead cells and suspended particles can also block or scatter light, limiting photon penetration in dense cultures.

Mechanical mixing can improve transport, but it may introduce shear stress from pumps or impellers that damages sensitive cells, membranes, or flagella depending on species' tolerance [102]. Therefore, reactor operation must balance improved mixing with cell integrity and long-term stability. Microgravity can also affect cell morphology and intracellular organization. Changes in chloroplast positioning, cytoplasmic streaming, and organelle orientation may reduce light-harvesting and carbon fixation [103]. These effects are often accompanied by shifts in gene expression, with more energy allocated to stress defense and repair rather than growth [104].

To address these challenges, specialized PBRs use capillary-driven flow, surface-tension effects, and microfluidic designs to improve nutrient and gas distribution under low-gravity [80, 105]. Computational fluid dynamics (CFD) is also valuable for optimizing reactor geometry, identifying stagnant zones, and reducing energy demand [106]. With appropriate engineering and strain selection, productive and stable microalgal systems can be developed for BLSS and space biomanufacturing during long-duration missions.

5.4.2 Cosmic radiation

Microalgae cultivated during long-duration missions are continuously exposed to GCR and SPEs, both of which can threaten cell survival and culture stability [16]. High-energy radiation can cause DNA damage, especially double-strand breaks, and generate ROS that harm proteins, lipids, and cell membranes [107]. Long-term exposure may increase mutation rates, genetic drift, and epigenetic changes, potentially reducing biomass productivity, altering nutritional quality, or triggering unwanted metabolite production. Protection requires combined biological and engineering strategies. Synthetic biology can improve radiation tolerance by enhancing DNA repair systems, antioxidant enzymes such as superoxide dismutase and catalase, and protective pigments that absorb UV or neutralize ROS [25, 30].

Physical shielding is also important. Water-rich PBRs can help attenuate charged particles, while dense algal biomass layers may provide additional passive protection [36, 108]. Natural pigments such as scytonemin and MAAs further support radioprotection through UV absorption and antioxidant activity [109]. As illustrated in **Figure 3**, metabolic pathways activated under radiation stress in microalgae

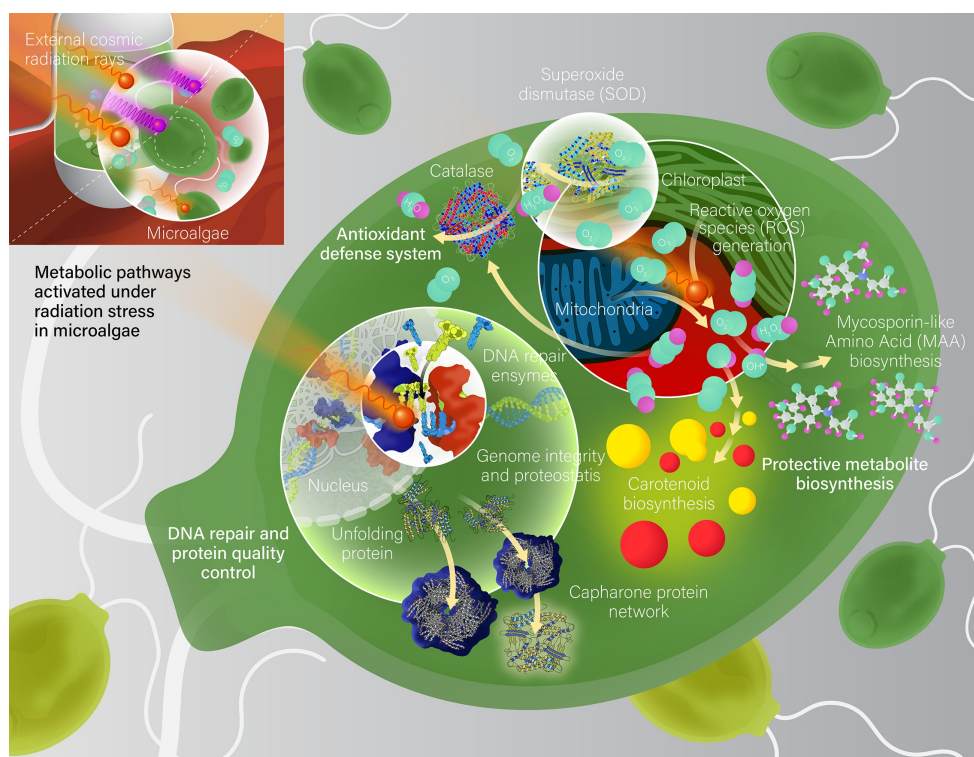


Figure 3.
Metabolic pathways activated under radiation stress in microalgae.

involve key stress-response mechanisms induced by exposure to cosmic radiation, including ROS generation, activation of antioxidant enzymes (e.g., superoxide dismutase and catalase), upregulation of carotenoid and MAA biosynthesis, and engagement of DNA repair and molecular chaperone protein networks.

Operational monitoring is equally critical. High-throughput sequencing, transcriptomics, metabolomics, and radiation dosimetry can detect harmful mutations or metabolic shifts early [40, 50]. Combined with adaptive control of light, nutrients, and reactor conditions, these strategies can maintain stable productivity under chronic radiation exposure. Together, strain engineering, shielding, and real-time management can help ensure reliable microalgal life-support performance in lunar, Martian, and deep-space habitats.

5.4.3 Resource limitations

Space-based microalgal cultivation is strongly limited by available mass, volume, water, and energy. These constraints are commonly evaluated using ESM, which combines mass, power, volume, cooling needs, and crew time into one mission-planning metric [16, 68]. Energy demand is one of the greatest challenges because lighting, temperature control, fluid circulation, and gas exchange consume significant power in PBR systems [110]. In space, waste heat is difficult to remove because convective cooling is absent. Therefore, efficient solutions such as pulsed

lighting, optimized light spectra, thermal insulation, and passive heat exchange are important for reducing energy use while maintaining productivity [111].

Water is another critical resource. Long-duration missions require highly efficient recycling systems, often targeting recovery rates above 99%, to minimize losses from leakage, evaporation, or incomplete separation [112, 113]. Water must be continuously reused across cultivation, hygiene, food production, and atmosphere control systems. Consumable components such as filters, sensors, pumps, and moving parts are also limited, so equipment must be durable, modular, and easy to repair. Additive manufacturing using algal-derived bioplastics may help produce replacement parts on demand and reduce reliance on Earth resupply [114].

To minimize system size and mass, productivity per unit volume must be maximized. This requires fast-growing strains and compact PBR designs such as flat-panel, capillary-driven, or membrane-based reactors that improve light delivery, gas exchange, and nutrient transport under microgravity [36]. Overall, sustainable microalgal BLSS will require integrated systems engineering that combines biological optimization, recycling efficiency, real-time control, and resource management to reduce dependence on Earth during long-duration missions.

5.4.4 Technical and biosafety issues

Maintaining contamination-free microalgal cultures in closed space habitats is a major challenge because nutrient-rich media can promote bacterial, fungal, viral, or phage growth that may destabilize cultures [115]. In BLSS, culture failure could disrupt oxygen production, CO₂ removal, nutrient recycling, and food supply, while recovery is limited by restricted crew time, energy, and consumables [36, 59]. Risk mitigation requires multilayered biosafety strategies, including sterilization, aseptic handling, and continuous microbial monitoring [116]. Real-time tools such as qPCR, flow cytometry, fluorescence assays, and sequencing can detect contamination early and enable rapid intervention [117].

Long-term cultivation also raises concerns over genetic drift and phenotypic instability caused by radiation, microgravity, and closed-system selection pressures, potentially reducing productivity or altering biomass quality [118]. Countermeasures include periodic reinoculation from cryopreserved master stocks, genetic containment systems, and continuous monitoring for harmful mutations or metabolic shifts [119]. Additional risks such as phage persistence, horizontal gene transfer, and cross-contamination highlight the need for reactor redundancy and operational isolation [120]. Overall, robust biosafety management is essential to ensure reliable microalgal cultivation and long-term BLSS stability in space.

6. AI, automation, and modeling for space algal systems

6.1 Predictive metabolic models

Predictive metabolic modeling is a key tool for optimizing microalgae under space conditions. Methods such as flux balance analysis (FBA) and flux variability analysis (FVA) simulate cellular mass and energy flows to predict growth and production of useful compounds such as lipids, astaxanthin, and phycocyanin [121, 122]. These models support strain selection and metabolic engineering by estimating

productivity, nutrient efficiency, and metabolite yields under constraints such as limited CO₂, radiation, and microgravity-related mass transfer issues [123, 124]. At the cellular level, FBA helps identify bottlenecks in carbon fixation, redox balance, and nitrogen assimilation under high-light or high-CO₂ cultivation conditions [25, 30, 121]. This information can guide synthetic biology strategies to redirect carbon toward lipids or antioxidant pigments, improve photoprotection, and reduce photorespiration, enabling high-performance strains for food, life-support, and biomanufacturing purposes. These models also improve cultivation strategies by optimizing batch, fed-batch, and continuous systems while minimizing nutrient inputs and maintaining gas-exchange efficiency [125]. When combined with real-time sensor data such as dissolved oxygen, CO₂ flux, nutrient levels, and biomass density, models can be continuously updated for adaptive control of PBR [126]. Overall, predictive metabolic modeling shifts algal cultivation from trial-and-error methods to intelligent, data-driven system design, supporting autonomous and resource-efficient BLSS for future long-duration space missions.

6.2 Digital twin systems for BLSS

Digital twin technology creates a high-fidelity virtual model of a BLSS that mirrors real algal cultivation systems in real time. By combining sensor data – such as pH, temperature, optical density, dissolved O₂/CO₂, and nutrient levels – with models including FBA, CFD, and growth kinetics, the system maintains an updated representation of reactor conditions [127–129]. This allows prediction of future culture states and early warning of issues such as oxygen oversaturation, CO₂ limitation, or nutrient depletion. Digital twins also improve anomaly detection by comparing expected and observed behavior. Deviations may indicate contamination, sensor drift, biofouling, gas-transfer problems, or hardware faults, enhancing biosafety and operational reliability where crew intervention is limited [130, 131].

In addition, digital twins enable virtual optimization and scenario testing. Algorithms such as genetic algorithms, simulated annealing, and reinforcement learning can optimize lighting spectra, photoperiods, mixing intensity, nutrient dosing, and harvest timing before changes are applied to the real system [132]. This reduces experimentation, saves energy and consumables, and lowers crew workload. Overall, digital twins transform BLSS from reactive systems into autonomous, self-regulating ecosystems capable of predictive maintenance, adaptive control, and stable oxygen generation, nutrient recycling, and biomass production for long-duration space missions.

6.3 Robotic maintenance

Automation and robotics are vital for maintaining long-term microalgal cultures in space while reducing crew workload and contamination risk [133]. Robotic systems integrated with PBRs can perform repetitive, high-precision tasks that are difficult or inefficient for astronauts [5]. These include automated harvesting and biomass separation through centrifugation or membrane filtration, allowing efficient biomass recovery while preserving optimal culture density. Robots can also handle cleaning and sterilization using UV treatment, ultrasonic cleaning, or controlled chemical wash cycles. These methods reduce biofouling, maintain light penetration, and preserve efficient heat and mass transfer, which is especially important in microgravity where manual cleaning is difficult [134].

In addition, robotic platforms can regulate nutrient delivery through precision pumps linked to sensor networks and digital twin systems. This allows accurate dosing that prevents nutrient deficiency, overfeeding, or metabolic stress [135]. Advanced robotic units equipped with manipulators, machine vision, and diagnostic sensors may also detect faults, replace damaged components, or isolate malfunctioning reactor modules without crew intervention [135, 136]. Overall, robotic maintenance improves consistency, productivity, and biosafety of algal systems. When combined with artificial intelligence (AI), real-time sensing, and digital twins, robotics enables highly autonomous and resilient BLSS for future missions to the Moon, Mars, and deep space.

6.4 Machine learning in PBR optimization

Machine learning (ML) is a powerful tool for optimizing PBR operation by handling the complex, nonlinear interactions that control microalgal growth [137]. Key variables such as light intensity and spectrum, temperature, pH, nutrient supply, hydrodynamic shear, and gas transfer interact in ways that often exceed the capabilities of conventional rule-based or PID controllers [138, 139]. The ML methods, especially artificial neural networks (ANNs) and deep learning models, can analyze large sensor datasets to identify patterns and predict culture performance [140]. A major application is dynamic light optimization, where ML models continuously adjust light intensity, photoperiod, and spectral quality to maximize photosynthesis while minimizing energy use [141]. This is especially valuable in space habitats, where lighting is a major contributor to ESM. Reinforcement learning can also optimize cultivation decisions over time under uncertain conditions while respecting safety constraints.

Machine learning further improves biosafety by enabling early contamination detection. Supervised and unsupervised models can recognize subtle changes in pH, dissolved oxygen, nutrient uptake, fluorescence, or cell morphology associated with bacterial, fungal, or phage invasion before culture collapse occurs [142]. ML can also distinguish true biological problems from sensor drift or equipment faults, reducing false alarms and unnecessary crew intervention. When integrated with digital twins, metabolic models, and robotic automation, ML enables adaptive and autonomous PBR management with stable biomass production, efficient nutrient recycling, lower energy demand, and rapid response to disturbances. These AI-driven systems are expected to be essential for reliable BLSS operation on lunar, Martian, and deep-space missions.

7. Future prospects and applications

7.1 Terraforming possibilities

Cyanobacteria and other phototrophic microorganisms may contribute to future planetary bioengineering or early terraforming strategies [15]. A natural precedent is Earth's Great Oxygenation Event, when ancestral cyanobacteria transformed the atmosphere through oxygenic photosynthesis over geological timescales [143]. Because some cyanobacteria tolerate low pressure, intense UV radiation, desiccation, and temperature extremes, they are considered potential candidates for early biological deployment on Mars or other worlds. In practice, initial use would likely

occur inside localized, pressurized, or semi-contained habitats rather than on open planetary surfaces. These controlled systems could optimize light, temperature, humidity, and nutrients to maximize photosynthesis and oxygen generation while protecting cells from harsh environmental conditions. Over long periods, such systems might gradually influence local atmospheres and serve as testbeds for planetary bioengineering concepts [144].

Microorganisms may also assist regolith transformation into biologically useful proto-soils. Cyanobacteria and microbial consortia release organic acids, siderophores, and chelating compounds that mobilize minerals such as phosphorus, potassium, magnesium, iron, and calcium [145]. Studies with Martian regolith simulants show that cyanobacteria can grow on these materials while enriching them with organic matter and bioavailable nutrients, helping prepare substrates for future plants or microbial communities [146]. Although planet-wide terraforming remains highly speculative and far beyond current capabilities, localized microbial systems could support oxygen production, nutrient cycling, and soil development. These approaches may help establish self-sustaining biospheres and reduce dependence on Earth resupply for long-term human settlement beyond Earth.

7.2 In situ resource utilization

Microalgae can become a key component of ISRU by helping produce and recycle essential resources directly on the Moon, Mars, or other planetary bodies [98]. Algal systems can be linked with local water extraction technologies that use subsurface ice or hydrated minerals, while also treating greywater and human waste through nutrient uptake and microbial remediation. This process can generate reusable water for crop production and life-support systems, reducing dependence on Earth-supplied water [1, 9, 147]. Microalgae may also support bioleaching, a low-energy method for recovering useful metals and minerals from planetary regolith [148]. In algal–bacterial consortia, microorganisms can solubilize elements such as iron, aluminum, magnesium, titanium, and rare-earth elements from lunar or Martian materials [149]. These recovered resources could be used for construction, radiation-shielding materials, 3D printing, electronics, and replacement parts.

When combined with nutrient recycling and carbon fixation, algal ISRU systems convert regolith into a productive resource base. Habitat CO₂ can be fixed into biomass, nitrogen and phosphorus can be recovered from wastes, and minerals can be mobilized from local substrates [10]. Together, these functions create an integrated closed-loop bioproduction platform. By embedding microalgae into ISRU strategies, future missions could lower resupply needs, reduce ESM, and increase operational autonomy. Such biologically driven systems offer sustainable pathways toward long-duration habitation and eventually self-sufficient extraterrestrial settlements.

7.3 Ultra-resilient engineered strains

Table 3 summarizes the engineering traits required for deep-space strains. Recent advances in synthetic biology and CRISPR–Cas genome editing have enabled the design of ultra-resilient microalgal and cyanobacterial strains for extreme space environments [25]. These engineered organisms can show improved resistance to ultraviolet and ionizing radiation through stronger DNA repair systems and increased production of protective pigments such as scytonemin, MAAs, and

Trait	Purpose	Engineering approach	Example outcomes
Radiation resistance	Protect DNA/proteins from cosmic rays	Overexpression of DNA repair enzymes	Reduced mutation rates
UV tolerance	Mitigate high UV exposure	Pigment biosynthesis enhancement	Increased scytonemin/MAAs
Desiccation tolerance	Enable dormant storage and transport	Trehalose accumulation, stress proteins	Survival under low moisture
Freeze–thaw resilience	Survive temperature fluctuations	Membrane stabilization, cold-shock proteins	Stable biomass production
High photosynthetic efficiency	Maximize O ₂ and biomass output	Optimized light-harvesting complexes	Improved CO ₂ fixation per unit area

Table 3.
Engineering traits required for deep-space strains.

carotenoids [25, 30]. These compounds help shield cells from radiation and oxidative stress while preserving photosynthesis. Engineered strains may also tolerate desiccation, osmotic stress, freeze–thaw cycles, and large temperature fluctuations. This can be achieved through accumulation of compatible solutes such as trehalose or glycine betaine, improved membrane stability, cryoprotectants, and stronger protein repair systems involving HSPs and molecular chaperones [150]. Such traits are important for long-term survival on the Moon, Mars, or deep-space platforms.

Beyond survival, metabolic engineering can improve productivity under low light, low temperature, or nutrient-limited conditions. Modified strains could efficiently produce single-cell protein (SCP) for food, lipids for biofuels or bioplastics, and bioactive compounds such as antioxidants, vitamins, and immunomodulatory molecules that support astronaut health [69]. Microalgae and cyanobacteria are also promising platforms for producing recombinant proteins and therapeutic compounds using only CO₂, light, and recycled nutrients [9]. In addition, biosafety remains essential. Genetic containment tools such as nutrient-dependent kill switches, inducible self-termination circuits, auxotrophic dependencies, and layered safeguards can reduce the risk of uncontrolled growth outside cultivation systems [9, 151]. These measures support both habitat safety and planetary protection. Overall, ultra-resilient engineered strains could become core technologies for space biomanufacturing, life-support systems, and sustainable long-duration missions beyond Earth.

7.4 Space biorefineries

Space biorefineries integrate microalgal cultivation with multifunctional product generation, supporting closed-loop life-support systems and long-term mission self-sufficiency [98]. Their goal is to maximize resource efficiency by converting most algal biomass into useful outputs while minimizing waste. Fresh biomass can provide SCP, essential amino acids, vitamins, and antioxidants, helping reduce nutritional deficiencies and muscle loss during extended missions [15, 98].

Lipid-rich biomass fractions may be converted into biodiesel, bioethanol, or biohydrogen for rovers, habitat systems, or auxiliary power, reducing reliance on Earth-supplied fuels [152]. Algal biopolymers such as polyhydroxyalkanoates (PHAs), cellulose derivatives, and extracellular polysaccharides can also be

processed into bioplastics for additive manufacturing. These materials could be used to 3D print tools, spare parts, structural items, and packaging on demand, lowering launch mass and increasing autonomy [15, 98]. Engineered strains may further produce antioxidants, vitamins, peptides, and other health-supporting compounds.

Beyond manufacturing, space biorefineries contribute directly to ecological closure. They consume crew-generated CO₂ and release oxygen through photosynthesis, functioning as biological air revitalization systems within BLSS [15]. They can also process greywater, inedible biomass, and human wastes, recycling nitrogen, phosphorus, and trace nutrients back into cultivation loops [15, 153].

The integration of AI-driven process control, real-time digital twins, automated robotics, and ultra-resilient engineered strains ensures that space biorefineries are scalable, adaptive, and resilient, capable of largely autonomous operation with minimal crew intervention [135]. This systems-level integration exemplifies the convergence of advanced biotechnology, systems engineering, and space science, positioning microalgae not merely as a source of food or oxygen, but as the foundational cornerstone of self-sustaining off-Earth habitats. By unifying resource recycling, food production, materials fabrication, and environmental stabilization, space biorefineries provide the technological and biological infrastructure necessary for deep-space settlements, lunar bases, and future Martian colonies. As illustrated in **Figure 4**, the algal-centered biomanufacturing ecosystem for Moon and Mars habitats depicts a multifunctional algal biorefinery designed for extraterrestrial

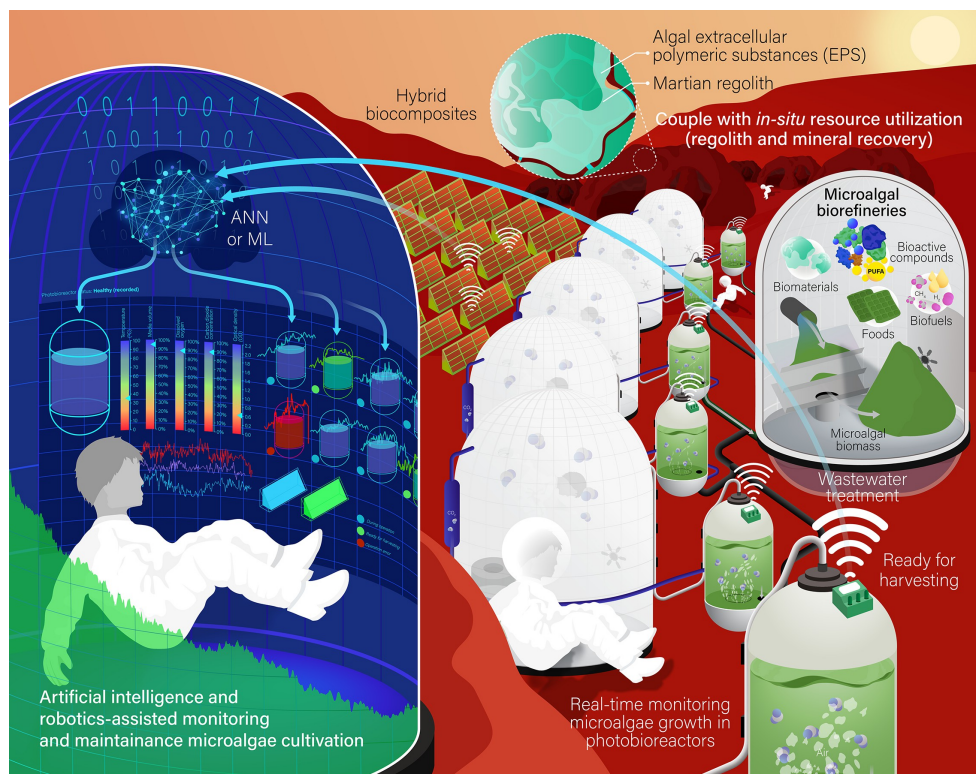


Figure 4.
 Algal-centered biomanufacturing ecosystem for Moon and Mars habitats.

environments, integrating water extraction and greywater recycling, the conversion of algal biomass into food, biofuels, pharmaceuticals, and biomaterials, coupling with ISRU processes (e.g., regolith processing and mineral recovery), and AI- and robotics-assisted system monitoring and maintenance.

8. Conclusions

Microalgae and cyanobacteria have emerged as key organisms for sustainable space exploration because they can simultaneously support air regeneration, waste recycling, food production, and bioproduct synthesis. Advances in PBR design, AI-based control systems, and engineered resilient strains have improved their readiness for integration into BLSS. Results from historical programs, ISS experiments, and omics studies confirm that these microorganisms can function under space-relevant conditions and may play major roles in long-duration missions and extraterrestrial settlements. However, important challenges remain. Microgravity and cosmic radiation create complex physiological stresses that are still not fully understood, especially during long missions. Uncertainty also remains regarding reactor scalability, large-volume cultivation, contamination control, and long-term ecological stability. In addition, integrated multi-species systems, closed-loop nutrient cycling, and ISRU concepts require further validation under autonomous operating conditions. To overcome these limitations, future research should prioritize long-duration, fully closed-loop BLSS experiments in orbit and in planetary analog habitats. These studies should evaluate not only algal productivity but also ecological interactions, genetic stability, and whole-system resilience over mission-relevant timescales. Integration of omics monitoring, AI-assisted modeling, digital twins, and robotic maintenance will further improve understanding, automation, and resource efficiency. Research should also focus on ultra-resilient engineered strains able to tolerate radiation, UV stress, desiccation, and temperature extremes while maintaining high productivity. At the same time, development of space biorefineries capable of producing food, fuels, medicines, and biomaterials from algal biomass will greatly enhance mission autonomy. Overall, with continued interdisciplinary research and system-scale validation, microalgae-based biotechnology can progress from experimental demonstrations to operational deployment, becoming a fundamental component of human life-support infrastructure beyond Earth.

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Conflict of Interest

The authors declare no conflict of interest.

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